

Competition Rules

Brake System Plausibility Device (BSPD)

- The BSPD is a standalone, non-programmable circuit.
- It functions to automatically open the shutdown circuit when hard braking occurs simultaneously with significant power delivery.
- For electric vehicles (EV), the BSPD triggers if 25 kW of power is delivered to the motors while the brakes are actuated.
- For internal combustion vehicles (CV), it triggers if the throttle position is more than 25% over the idle position during hard braking.
- The device must take action to open the shutdown circuit if this implausibility condition persists for more than 500ms.
- To detect hard braking, the system must utilize a brake system pressure sensor.
- The importance of the BSPD lies in its ability to force the vehicle into a safe state by cutting power if simultaneous acceleration and braking are detected, preventing runaway acceleration in the event of a throttle or motor control failure.

Accumulator Isolation Relay (AIR)

- At least two AIRs must be fitted inside each Tractive System Accumulator Container (TSAC).
- AIRs must be mechanical relays of a "normally open" type.
- Solid-state relays are strictly prohibited for this function.
- The functionality of the AIRs is to open both poles of the TS accumulator.
- The importance of the AIRs is that when they are open, no TS voltage may be present outside of the TSAC.
- This mechanism ensures that the vehicle side of the AIRs is galvanically isolated from the accumulator side, protecting the rest of the vehicle and personnel from high voltage.

Brake Over-Travel Switch (BOTS)

- The BOTS is a mechanical single pole, single throw switch.
- It is integrated into the vehicle as a direct part of the shutdown circuit.
- It is installed so that a failure in at least one of the brake circuits causing brake pedal over-travel will result in the shutdown circuit opening.

- Repeated actuation of this switch must not close the shutdown circuit.
- It is designed so that the driver cannot manually reset it from the cockpit.
- The importance of the BOTS is to act as a critical mechanical failsafe that cuts all vehicle power if there is a catastrophic loss of hydraulic braking pressure.

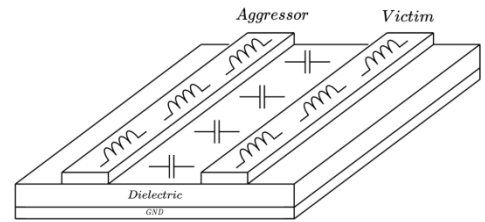
Inertia Switch

- The inertia switch is integrated into the shutdown circuit so that an impact will result in the circuit being opened.
- The device must trigger due to an omnidirectional peak acceleration of ≤ 8 g for a half sine test pulse of 250ms length.
- It must also trigger at ≤ 13 g for a half sine test pulse of ≥ 20 ms length.
- The inertia switch must securely latch until it is manually reset.
- The device must not include any semiconductor components.
- This switch is important because it physically cuts the vehicle's power during a crash or collision without relying on complex software, mitigating post-impact fire or electrocution risks.

PCB & Circuits

2. What is Crosstalk in PCB and How to reduce it?

crosstalk is the unwanted transfer of energy from one element (the aggressor) to another element (the victim). These elements can be traces, components, planes, and so on.

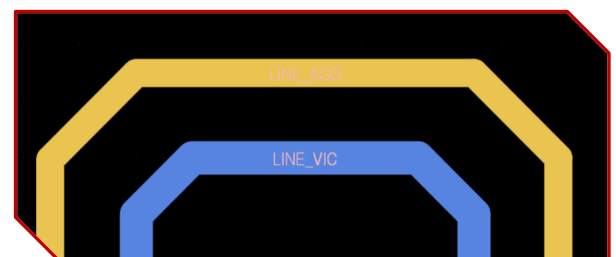


This energy transfer happens due to capacitive and inductive coupling, as all real-world elements have some amount of parasitic and stray capacitances and inductances associated with them. For crosstalk to occur, we then require that we have changing signals (voltages and current) over time.

(How to reduce it)

1-Spacing

Increasing the physical spacing between the aggressor and victim traces is one of the most effective methods to minimize crosstalk in PCB design.



2- Decrease Distance to Reference Planes

Decreasing the distance between signal traces and reference planes reduces crosstalk by tightly containing the electromagnetic fields.

1	L1 (Sig)	CF-004	Signal	1oz	0.035mm
	PP1	PP-017	Prepreg		0.21mm
2	L2 (GND)	CF-004	Plane	1oz	0.0152mm

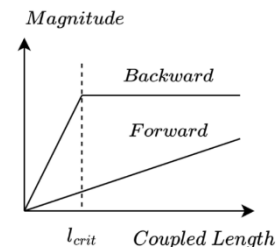
3-Avoid Adjacent Signal Layers

Avoid placing signal layers adjacent to each other, or use orthogonal routing (horizontal on one layer, vertical on the other) if they must be adjacent.

1	L1 (Sig)
	PP1
2	L2 (GND)
	C1
3	L3 (GND)
	PP2
4	L3 (Sig)

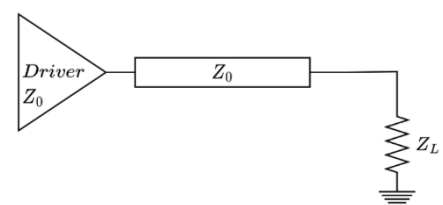
4-Reducing Lengths of Parallel Routes

Minimizing the length of parallel traces can reduce crosstalk, though its effectiveness is often limited since backward crosstalk dominates even at short lengths.



5 – Termination

Proper signal termination can mitigate crosstalk by reducing high-frequency ringing, but it requires careful calculation and simulation before application.



3. Build this circuit in your simulator and answer

a) What is this circuit's primary function?

an Active-High digital input, using a pull-down resistor to prevent floating states and filter noise.

b) What do the components inside the dashed box represent? Explain in detail.

represent environmental electromagnetic interference (EMI) coupling into the circuit:

- The 120V 50Hz AC source simulates ambient interference from nearby mains power lines.
- The massive 1G resistor (R3) models the high-impedance parasitic coupling (like stray capacitance through the air) between the noise source and our unshielded wire, showing it's a weak electromagnetic connection rather than a direct physical short.

c) R1 (the main character)

1) What happens if you remove R1? (Simulate the result and explain).

The pin becomes highly susceptible to noise. Without R1, the AC noise voltage drops across the massive 100M internal resistance, causing the input to read fluctuating, artificially high voltages instead of a clean 0V.

2) What is an input called when a resistor like R1 is missing?

A floating pin

3) How should the resistance value for R1 be selected in a real circuit?

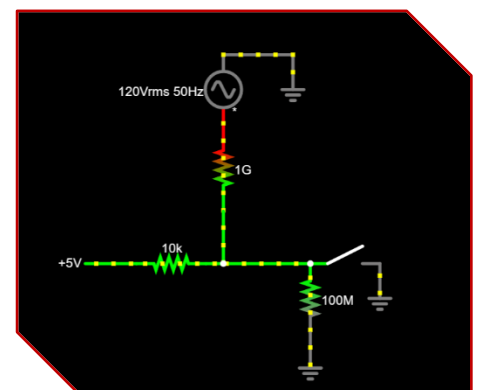
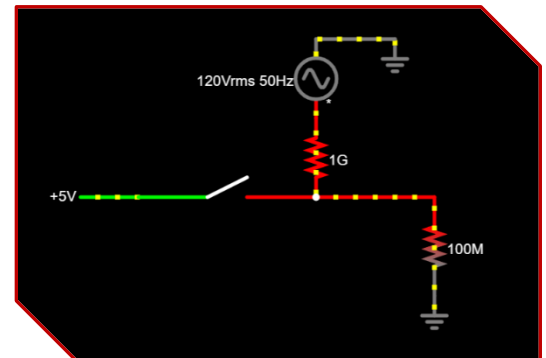
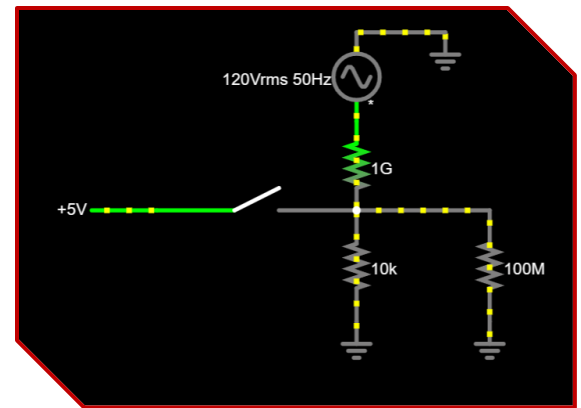
The value is a trade-off: low enough to filter noise effectively, but high enough to prevent battery drain. Typical values range between $1\text{k}\Omega$ and $10\text{k}\Omega$.

d) What is the arrow likely pointing to, and what role does R2 serve? What could we call R2?

The arrow points to the microcontroller input pin. R2 represents the microcontroller's extremely high internal resistance, allowing it to sense voltage levels without drawing significant current from the circuit. We call R2 the Internal Input Impedance.

e) If our active signal voltage were 0V instead of 5V, how should the circuit be modified? What is this new circuit topology called?

The switch must be connected to ground (0V), and the resistor must be connected to the +5V supply. This new topology is called a Pull-up resistor circuit (which functions as an Active-Low input).



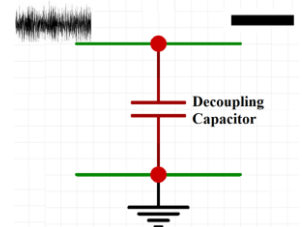
f) What would happen if you brought your hands close to (or touched) an unshielded question)

The human body acts as an antenna that picks up environmental electromagnetic noise (like 50Hz AC mains). Bringing your hand close increases the parasitic capacitance, injecting this noise into the unshielded circuit. If the pin is floating, this will cause severe interference and false digital readings (similar to the buzzing sound when touching an exposed audio cable).

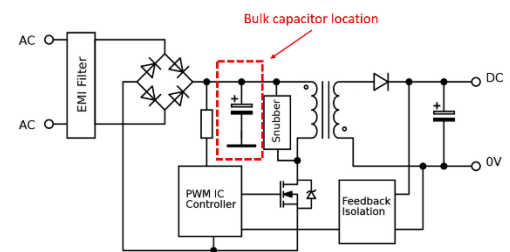
4.Explain the difference between Decoupling Capacitors and Bulk Capacitors and when we use them.

Decoupling Capacitors are placed as close as possible to the power pins of individual ICs. They act as local energy reserves to stabilize voltage and supply instant current during high-speed switching. They are typically small ceramic capacitors ($0.1\mu\text{F}$ to $1\mu\text{F}$) used to handle medium-frequency changes.

Bulk Capacitors are placed at the main power entry point or near voltage regulators. They act as large energy reservoirs for the entire circuit, smoothing out low-frequency voltage variations and compensating for sudden, massive power demands. They are typically large electrolytic or tantalum capacitors ($10\mu\text{F}$ to $1000\mu\text{F}$).



Decoupling Capacitor



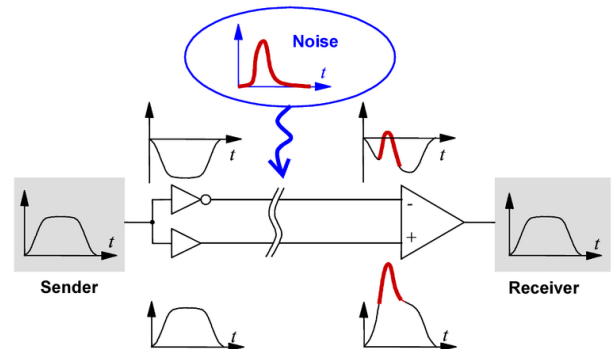
Bulk Capacitor

Point	Decoupling Capacitors	Bulk Capacitors	Bypass Capacitor (Bonus)
Main Role	Stabilizes voltage and supplies instant current to the IC (local backup battery).	Stores large energy to compensate for sudden current demands (main reservoir).	Shunts high-frequency noise to ground before it affects the signal.
Placement	As close as possible to the IC's power pins.	At the power entry or next to the voltage regulator.	Across VCC and GND lines near noisy ICs.
Typical Value	$0.1\mu\text{F}$ to $1\mu\text{F}$	$10\mu\text{F}$ to $1000\mu\text{F}$	$0.01\mu\text{F}$ to $0.1\mu\text{F}$
Material/Type	Ceramic	Electrolytic or Tantalum	Ceramic (Low ESR & ESL)
Target Frequency	Medium frequency	Low frequency	High frequency

5. What are Differential Pairs?

In PCB design, a differential pair is a transmission line topology consisting of two closely coupled traces used to transmit a single data signal. Instead of referencing the signal to ground (single ended transmission), the driver outputs two complementary signals equal in amplitude but opposite in phase (D+ and D-). The receiver reconstructs the data by calculating the voltage difference between the two. The primary advantage of this method is its high immunity to electromagnetic interference (EMI).

Any external noise typically couples onto both traces equally (common-mode noise) and is inherently canceled out by the receiver's differential amplifier.



Critical Rules for Routing Differential Pairs:

1. **Length Matching (Intra-pair Skew):** Both traces must be exactly the same length. Any length mismatch introduces a phase shift (skew), meaning the signals won't arrive at the receiver simultaneously. This degrades signal integrity and ruins the common-mode noise cancellation.
2. **Constant Spacing and Width:** The gap between the two traces, along with their trace width, must remain strictly constant throughout the entire route. Any variation in this spacing alters the differential impedance (e.g., 90Ω for USB), which leads to signal reflections and losses.
3. **Routing Symmetry:** The physical layout should be as symmetrical as possible. If a via or a bend is unavoidable on one trace, it should ideally be mirrored on the other. Bends should be routed using 45-degree angles or smooth arcs to prevent impedance discontinuities.
4. **Clearance (Isolation):** Differential pairs must be properly isolated from other noisy signals, such as clocks or power switching nodes, to prevent crosstalk. A standard practice is to apply the "3W rule," maintaining a clearance distance of at least three times the trace width from other adjacent nets.

6.1. What are Polygon Pours?

In PCB design, a polygon pour is a multi-sided area of copper placed on a signal layer, which can be configured as either a solid fill or a hatched pattern. The primary advantage of a polygon pour is its intelligent behavior; it automatically pours around and avoids other copper objects (such as pads, vias, or tracks) that belong to a different net. It does this while strictly following the defined Electrical Clearance and Polygon Connect Style Design Rules.



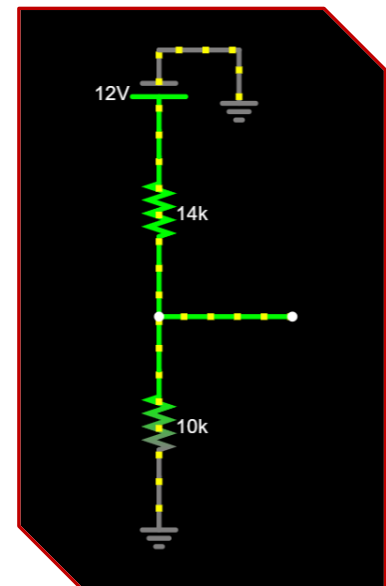
6.2. Why are they preferred in Power and GND connections?

- **Carrying Large Currents (Power):** Utilizing a solid polygon pour for power connections creates a large copper area that significantly reduces electrical resistance. This is crucial for efficiently carrying large power supply currents without experiencing excessive voltage drops or thermal issues.
- **Electromagnetic Shielding (GND):** Pouring a ground-connected polygon over the unused areas of a signal layer provides essential electromagnetic shielding. It establishes an electrically-stable ground reference throughout the PCB, minimizing noise and preventing crosstalk between sensitive signal traces.
- **Automated Clearance Management:** Because polygon pours automatically calculate voids and clearances around foreign nets, they allow designers to safely maximize the copper coverage for Power and GND networks. This eliminates the risk of accidental short circuits and saves significant time compared to complex manual routing.

a) Why have you chosen these specific values for your components?

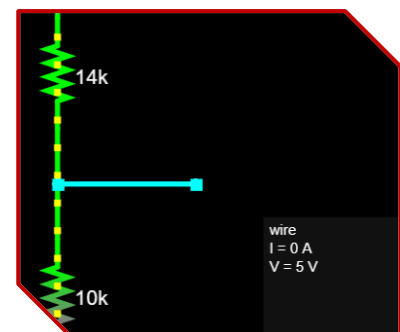
To meet both the voltage step-down requirement and the power budget simultaneously:

1. **Power Budget:** $P = \frac{V_{in}^2}{R_{total}} \Rightarrow 0.006 \text{ W} = \frac{12^2}{R_{total}} \Rightarrow R_{total} = 24 \text{ k}\Omega$
2. **Voltage Divider:** $V_{out} = V_{in} \cdot \left(\frac{R_2}{R_{total}}\right) \Rightarrow 5\text{V} = 12\text{V} \cdot \left(\frac{R_2}{24\text{k}\Omega}\right) \Rightarrow R_2 = 10 \text{ k}\Omega$



b) What would happen to the output of your circuit?

The leakage current acts as a load, dropping the output voltage significantly because our divider has a relatively high Thevenin resistance.

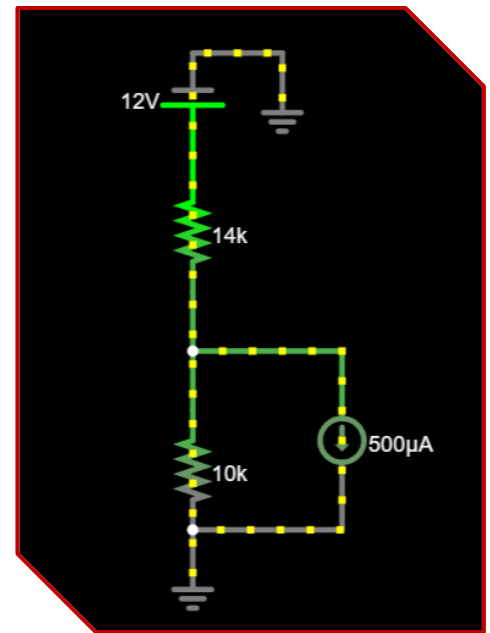


- **Thevenin Resistance (R_{th}):** $R_{th} = 14\text{ k}\Omega \parallel 10\text{ k}\Omega = 5.833\text{ k}\Omega$
- **Voltage Drop:** $V_{drop} = I_{leakage} \cdot R_{th} = 0.5\text{ mA} \cdot 5.833\text{ k}\Omega = 2.916\text{ V}$
- **New Output:** $V_{out(new)} = 5\text{ V} - 2.916\text{ V} = 2.084\text{ V}$ (The signal is severely corrupted).

c) What is the maximum acceptable leakage current?

Assuming a standard 10-bit ADC, the maximum acceptable voltage drop should be less than 1 LSB to prevent losing signal accuracy.

- **1 LSB:** $\frac{5\text{ V}}{1024} = 4.88\text{ mV}$
- **Max Leakage:** $I_{max} = \frac{V_{drop_max}}{R_{th}} = \frac{4.88\text{ mV}}{5.833\text{ k}\Omega} \approx 0.836\text{ }\mu\text{A}$



8. What will happen if two interrupt requests arise at the same time?

When two interrupt requests occur simultaneously, the microcontroller evaluates their assigned priority levels. The interrupt with the higher priority (lower numerical value) is serviced first, while the other is placed in a pending state. *Bonus (ARM Context)*: In ARM Cortex-M processors, this is managed by the Nested Vectored Interrupt Controller (NVIC). The NVIC supports priority grouping, allowing interrupts to have a Preemption Priority and a Sub-Priority. If two simultaneous interrupts have the same Preemption Priority, the NVIC uses the Sub-Priority (or the hardware exception number) to break the tie.

10. What are communication protocols in the context of embedded systems? Provide at least Three examples.

Communication protocols are formal sets of rules and conventions that govern how microcontrollers, sensors, and external peripherals exchange digital data securely and reliably. They define pin configurations, timing, data framing, and error checking. *Examples*:

1. **I2C (Inter-Integrated Circuit)**: A synchronous, multi-master, multi-slave protocol utilizing two wires (SDA and SCL). It is ideal for short-distance, low-speed communication with multiple sensors.
2. **SPI (Serial Peripheral Interface)**: A synchronous, full-duplex protocol utilizing four wires (MOSI, MISO, SCK, CS). It is suited for high-speed data transfer, such as interfacing with SD cards or displays.
3. **UART (Universal Asynchronous Receiver-Transmitter)**: An asynchronous, point-to-point hardware protocol utilizing two wires (TX and RX). It is commonly used for PC-to-MCU debugging and long-distance serial communication.

11. What are the differences between the following families and their use cases: STM32F4, Raspberry Pi Zero, ATmega.

- **ATmega (e.g., ATmega328P)**: 8-bit AVR microcontrollers running at low clock speeds (e.g., 16 MHz).
 - *Use Case*: Simple, low-power applications like basic sensor polling, simple automation, or introductory DIY projects.
- **STM32F4**: 32-bit ARM Cortex-M4 microcontrollers with hardware floating-point units (FPU), running at high speeds (up to 180 MHz).
 - *Use Case*: Real-time, compute-intensive hardware applications such as complex motor control algorithms, digital signal processing (DSP), and fast sensor fusion.
- **Raspberry Pi Zero**: A complete single-board computer (SBC) featuring a 32-bit ARM processor (1GHz) capable of running a full Linux operating system.
 - *Use Case*: Applications requiring OS management, network connectivity, high-level programming (Python/C++), and heavy data processing like computer vision or IoT gateways.

GitHub Repo : [Omarr-Darwish/LV Solo Mission](#)